

DYLLAN BROWN

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PROFESSIONAL SUMMARY

Mechanical Design Engineer with 5+ years of hands-on experience taking parts and equipment from concept to production — combining SolidWorks/AutoCAD design work, GD&T and DFM/DFMEA analysis, rapid prototyping, and Design-of-Experiments validation with strong root-cause and reliability engineering instincts. Experience spans both production equipment design and new part/product development within a high-volume, highly automated manufacturing environment. Looking to bring that concept-through-validation design skill set to a Design Engineer role in either equipment design or product development.

WORK EXPERIENCE

Senior Manufacturing Equipment Engineer

June 2021 to Present

Tesla, Inc – 4680 Battery Cell Manufacturing – Austin, Texas

Equipment Design & Improvement

- Reverse-engineered spare parts for supported tools to work around a key vendor's extended lead times — converting vendor 2D drawings into 3D SolidWorks models, then producing STEP files and GD&T drawings for manufacture by third-party machine shops.
- Redesigned hard-stop fault logic and engineered fixed-offset limits with automated alerting for clutch position sensors — raising machine availability from 83% to 97%+, cutting clutch trip occurrences 84%, and closing a recurring sensor mis-calibration risk, for a fleet-wide +67 ppm gain in Demonstrated Capability.
- Designed and deployed a support modification for a CIP subassembly to eliminate a cantilever deflection issue causing chronic chemical leaks — improving equipment availability 12% and capacity 5%, eliminating 2,000+ monthly maintenance dispatch hours, and cutting downstream quality fallout ~10%; now a standard commissioning requirement across all electrolyte fillers.
- Applied DFM principles to redesign an existing multi-piece assembly into a single-piece design, producing new GD&T-toleranced drawings for the part and deploying the upgrade — eliminating a chronic chemical leak, cutting annual spare parts cost ~\$10K, and eliminating a dented-terminal quality escape.
- Designed and validated (via Design of Experiments) a breakaway torque change on a continuous-motion turret, eliminating recurring ghost trips that had extended individual reset times by 20+ minutes while maintaining strict quality requirements.
- Identified the root cause of a recurring yield reject and designed an equipment upgrade that was deployed fleet-wide — sustaining above 99.5% yield and contributing to a 75% reduction in total scrap cost across the line for the year.
- Led design and commissioning of a new production tool and a dedicated equipment rebuild space — including process layout, equipment design, and supporting facility utilities — validating process and equipment capabilities ahead of launch.

Equipment Reliability & Process Support

- Delivered a 65 ppm increase in average Demonstrated Capability across 4680 build programs through equipment design upgrades, controls logic improvements, and a new preventative maintenance program built to sustain long-term reliability — while eliminating dented-cell containments.
- Served as onsite engineering lead for OEM equipment upgrades — deploying ECN design changes to improve equipment performance and reliability.
- Designed and led a first-of-its-kind equipment overhaul procedure for a critical production tool with no existing repair process — successfully completing the overhaul and establishing the maintenance instruction now used for future overhauls.
- Led production readiness and commissioning activities across multiple equipment areas for a new production line, enabling a doubling of production capacity for a new cell variant.
- Correlated barometric pressure trends with incoming weather patterns to diagnose sudden vacuum supply pressure shifts across a fleet of production equipment, then designed and rolled out an equipment upgrade fleet-wide to resolve the issue.
- Designed and built several real-time equipment health metric dashboards to accelerate floor troubleshooting and reduce equipment downtime.

Process Engineer

July 2019 to June 2021

Anheuser-Busch InBev – Columbus, Ohio

- Supported plant operations in process, production, project, and maintenance engineering roles; directly overseeing critical equipment in the Packaging Department.
- Improved filler and closer equipment uptime by 5% in first year through maintenance execution and root cause analysis problem solving.
- Developed and deployed maintenance routines around equipment areas to ensure performance and improve equipment uptime.
- Developed and deployed Standard Operating Procedures for preventative maintenance routines, production reaction plans, and equipment center conversions.
- Managed parts procurement, service contracts, and timeline planning for several large six-figure CapEx maintenance overhauls.

Operations Group Manager

February 2018 to July 2019

Anheuser-Busch InBev – Columbus, Ohio

- Supervised 7 packaging lines and over 30 employees daily across all three shifts.
- Owned line performance and ensured daily volume demands were met.
- Analyzed line data trends to identify and close performance gaps.
- Prioritized tasks and made critical decisions impacting six figures worth of product each day.

SKILLS

- SolidWorks
- SolidWorks PDM
- AutoCAD
- GD&T
- DFM
- DFMEA / PFMEA
- Mechanical Design & Testing
- Equipment Installation & Commissioning
- DOE & Statistical Analysis
- JMP
- Data Visualization & Dashboard Development in Grafana
- Custom Data Querying
- Root Cause Analysis
- MES Systems
- Six Sigma / Continuous Improvement
- Vision Software & Hardware
- Project Management
- PLC / Controls (Beckhoff TwinCAT, Structured Text)
- 3D Printing

EDUCATION

Bachelor of Science, Mechanical Engineering

December 2017

The University of Toledo – Toledo, Ohio | GPA: 3.511 | Graduated Cum Laude

INTERESTS

- Music Production (Ableton Live)
- Sound Design
- Audio Engineering, Mixing & Mastering